

习题:

设 X 表示10次独立重复射击命中目标的次数, 每次射中目标的概率为0.4, 则 X^2 的数学期望 $EX^2 =$ _____.

解 $X \sim B(10, 0.4)$

$$EX = np = 10 \times 0.4 = 4$$

$$DX = npq = 10 \times 0.4 \times 0.6 = 2.4$$

$$EX^2 = DX + (EX)^2 = 18.4$$

习题:

设随机变量 X 的分布函数 $F(x)$ 为

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{2} + x + \frac{x^2}{2} & -1 \leq x < 0 \\ \frac{1}{2} + x - \frac{x^2}{2} & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases} \quad \text{求} EX \text{和} DX.$$

解

$$f(x) = F'(x) = \begin{cases} 1+x, & -1 < x < 0 \\ 1-x, & 0 \leq x < 1 \\ 0, & x < -1 \text{ 或 } x \geq 1 \end{cases}$$

$$f(x) = F'(x) = \begin{cases} 1+x & -1 < x < 0 \\ 1-x & 0 \leq x < 1 \\ 0 & \text{其他} \end{cases}$$

$$EX = \int_{-\infty}^{+\infty} xf(x)dx$$

$$= \int_{-1}^0 x(1+x)dx + \int_0^1 x(1-x)dx = 0$$

$$EX^2 = \int_{-\infty}^{+\infty} x^2 f(x)dx$$

$$= \int_{-1}^0 x^2(1+x)dx + \int_0^1 x^2(1-x)dx = \frac{1}{6}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{6}$$

习题:

设 X 服从参数为1的指数分布, 则 $f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$;
 $DX^2 = \underline{\hspace{2cm}}$.

$$\text{解 } \because X \sim E(1) \quad \therefore EX = DX = 1$$

$$\begin{aligned} E(e^{-2X}) &= \int_{-\infty}^{+\infty} e^{-2x} f(x) dx = \int_0^{+\infty} e^{-2x} e^{-x} dx \\ &= \int_0^{+\infty} e^{-3x} dx = -\frac{1}{3} e^{-3x} \Big|_0^{+\infty} = \frac{1}{3} \end{aligned}$$

$$EX^2 = DX + (EX)^2 = 1 + 1^2 = 2$$

$$EX^4 = \int_{-\infty}^{+\infty} x^4 f(x) dx = \int_0^{+\infty} x^4 e^{-x} dx = 24$$

$$DX^2 = EX^4 - (EX^2)^2 = 24 - 2^2 = 20$$

习题：

设随机变量 $X \sim N(2, \sigma^2)$, 且 $P\{2 < X < 4\} = 0.3$,

则 $P\{X < 0\} = \underline{\hspace{2cm}}$.

$$\begin{aligned}\text{解 } P\{2 < X < 4\} &= F(4) - F(2) = \Phi\left(\frac{4-2}{\sigma}\right) - \Phi\left(\frac{2-2}{\sigma}\right) \\ &= \Phi\left(\frac{2}{\sigma}\right) - \frac{1}{2} = 0.3 \Rightarrow \Phi\left(\frac{2}{\sigma}\right) = 0.8\end{aligned}$$

$$\text{于是 } P\{X < 0\} = F(0) = \Phi\left(\frac{0-2}{\sigma}\right) = 1 - \Phi\left(\frac{2}{\sigma}\right) = 0.2$$

作业5: 已知 $X \sim f(x) = \begin{cases} \frac{1}{2} \cos \frac{x}{2}, & 0 < x < \pi, \\ 0, & \text{others.} \end{cases}$

对 X 独立观察 4 次, Y 表示 X 的观察值大于 $\pi/2$ 的次数,
求 $E(Y^2)$.

解: 由题意可知 $Y \sim b(4, p)$

$$p = P\{X > \frac{\pi}{2}\} = \int_{\pi/2}^{\pi} \frac{1}{2} \cos \frac{x}{2} dx = 1 - \frac{\sqrt{2}}{2}$$

$$\begin{aligned} E(Y^2) &= D(Y) + E^2(Y) = np(1-p) + (np)^2 \\ &= 22 - 14\sqrt{2} \end{aligned}$$